Section 02: Mathematical Ability

Module 15: YouTube Video Lectures & Marathon PYQ Master Vault (Infosys Systems Engineer)

> **Document Metadata & Provenance:**
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> - **Source Corpus:** Authentic Infosys Systems Engineer (SE) YouTube Video Lectures, Live Slot Simulations & Assessment Analyses.
> - **Primary Channels:** OnlineStudy4U (Pratik Sir), CampusMonk (Rachit Sir), KG Placement (Yash Jain Sir), CareerRide.
> - **Curriculum Target:** Section 02: Mathematical Ability & Data Interpretation (Tables, Bar Graphs, Pie Charts, Pipes & Cisterns, Speed Time & Distance, Probability, Number System).
> - **Total Questions:** 45 Forensic Questions with complete step-by-step mathematical proofs, algebraic equations, and KaTeX derivations.
> - **Zero Placeholders:** Complete dataset tables, formula applications, and shortcut tricks.

Question 1

Source: OnlineStudy4U — *Infosys & LTM Aptitude Mock Test | Data Interpretation (Pratik Sir)* The following table shows the export value (in Crore INR) of four companies (A, B, C, D) in the years 2024 and 2025:

Company	2024 Export (₹ Cr)	2025 Export (₹ Cr)
A	100	130
B	120	150
C	150	180
D	80	110

Which company recorded the highest percentage growth in exports from 2024 to 2025?

- [A] Company A
- [B] Company B
- [C] Company C
- [D] Company D

Correct Answer: [D] Company D

Step-by-Step Derivation & Mathematical Proof:

1. **Growth Formula:** ext{Percentage Growth} = frac{ ext{Export}_{2025} - ext{Export}_{2024}}{ ext{Export}_{2024}} imes 100%
2. **Calculate for each company:**
   - **Company A:** ext{Growth}_A = frac{130 - 100}{100} imes 100% = frac{30}{100} imes 100% = 30.00%
   - **Company B:** ext{Growth}_B = frac{150 - 120}{120} imes 100% = frac{30}{120} imes 100% = frac{1}{4} imes 100% = 25.00%
   - **Company C:** ext{Growth}_C = frac{180 - 150}{150} imes 100% = frac{30}{150} imes 100% = frac{1}{5} imes 100% = 20.00%
   - **Company D:** ext{Growth}_D = frac{110 - 80}{80} imes 100% = frac{30}{80} imes 100% = frac{3}{8} imes 100% = 37.50%

3. **Comparison:**

$$37.50\%(extD) > 30.00\%(extA) > 25.00\%(extB) > 20.00\%(extC)$$

4. **Key Exam Shortcut:**
   - Notice that the absolute increase for all four companies is identical: exactly ₹$30extCrore$!
   - When absolute increase ΔX is constant, percentage increase $racΔXX_0$ is strictly **maximized when the base X_0 is minimized**!
   - Since Company D has the smallest initial base (80), its percentage growth is mathematically guaranteed to be the highest.
5. Therefore, **Company D** recorded the highest percentage growth (37.5%).

Question 2

Source: OnlineStudy4U — *Infosys 1SHOT Aptitude (Pratik Sir)* Two pipes, A and B, can fill a cistern when operating together in 1↑rac{7}{8} hours (i.e., $rac158$ hours). If opened separately, Pipe B takes 2 hours more than Pipe A to fill the cistern. Find the time taken by Pipe A alone to fill the cistern.

- [A] 3 hours
- [B] 4 hours
- [C] 5 hours
- [D] 6 hours

Correct Answer: [A] 3 hours

Step-by-Step Derivation & Mathematical Proof:

1. **Define Variables:**
   - Let the time taken by Pipe A alone be x hours.
   - Then Pipe B takes $(x + 2)$ hours.
2. **Work-Rate Formulation:** ext{Rate of A} = ↑rac{1}{x}, \quad ext{Rate of B} = ↑rac{1}{x + 2} ext{Combined Rate} = ↑rac{1}{x} + ↑rac{1}{x + 2} = ↑rac{1}{15/8} = ↑rac{8}{15}
3. **Algebraic Simplification:** rac{(x + 2) + x}{x(x + 2)} = ↑rac{8}{15} rac{2x + 2}{x^2 + 2x} = ↑rac{8}{15} \implies ↑rac{2(x + 1)}{x(x + 2)} = ↑rac{8}{15} rac{x + 1}{x(x + 2)} = ↑rac{4}{15}

$$15(x + 1) = 4(x^2 + 2x)$$

$$15x + 15 = 4x^2 + 8x$$

$$4x^2 - 7x - 15 = 0$$

4. **Solve the Quadratic Equation:**

$$\Delta = (-7)^2 - 4(4)(-15) = 49 + 240 = 289 = 17^2$$

x = ↑rac{7 \pm 17}{2 imes 4} = ↑rac{7 \pm 17}{8}

   - Positive root (since time cannot be negative): x = ↑rac{7 + 17}{8} = ↑rac{24}{8} = 3 ext{ hours}
5. **Verification:**
   - A takes 3 hours (Rate = $rac13$).
   - B takes $3 + 2 = 5$ hours (Rate = $rac15$).
   - Combined rate: rac{1}{3} + ↑rac{1}{5} = ↑rac{5 + 3}{15} = ↑rac{8}{15}.
   - Combined time: $rac158 = 1.875exthours = 1exthr52.5extmin$. Exact match!

6. Pipe A alone takes **3 hours**.

Question 3

Source: OnlineStudy4U — *Infosys System Engineer PYQ Most Important Topics (Pratik Sir)* A bag contains 5 red balls, 4 green balls, and 3 blue balls. If two balls are drawn at random without replacement, what is the probability that both balls are of different colors?

- [A] $rac4766$
- [B] $rac1966$
- [C] $rac2333$
- [D] $rac3166$

Correct Answer: [A] $rac4766$

Step-by-Step Derivation & Complement Rule:

1. **Total Number of Balls:**

$$N = 5 + 4 + 3 = 12 ext{balls}$$

2. **Total Possible Outcomes:** Number of ways to choose 2 balls out of 12: □inom{12}{2} = □rac{12 imes 11}{2} = 66
3. **Use the Complement Rule:**

$$\mathbf{P}(ext{Both Different}) = 1 - \mathbf{P}(ext{Both Same Color})$$

   - Both Red: □inom{5}{2} = □rac{5 imes 4}{2} = 10
   - Both Green: □inom{4}{2} = □rac{4 imes 3}{2} = 6
   - Both Blue: □inom{3}{2} = □rac{3 imes 2}{2} = 3
   - Total ways to draw 2 balls of the same color:

$$ ext{Same Color Ways} = 10 + 6 + 3 = 19$$

4. **Probability Calculation:** \mathbf{P}(ext{Both Same Color}) = ↑rac{19}{66} \mathbf{P}(ext{Both Different}) = 1 - ↑rac{19}{66} = ↑rac{66 - 19}{66} = ↑rac{47}{66}
5. Result is $rac4766$ [A].

Question 4

Source: KG Placement — *1.4: Infosys Aptitude - Speed Time & Distance (Yash Jain Sir)* A train 240 meters long crosses a platform of length 360 meters in 30 seconds. What is the speed of the train in km/h?

- [A] 64 km/h
- [B] 72 km/h
- [C] 80 km/h
- [D] 90 km/h

Correct Answer: [B] 72 km/h

Step-by-Step Derivation & Unit Conversion:

1. **Total Distance Traversed:** When a train completely crosses a platform, the total distance covered equals the sum of the lengths of the train and the platform:

$$D = L_{ ext{train}} + L_{ ext{platform}} = 240 ext{m} + 360 ext{m} = 600 ext{meters}$$

2. **Speed in m/s:** v = ↑rac{D}{t} = ↑rac{600 ext{ m}}{30 ext{ s}} = 20 ext{ m/s}

3. **Convert m/s to km/h:** Multiply by the conversion factor $rac185$: v_{ ext{km/h}} = 20 imes rac{18}{5} = 4 imes 18 = 72 ext{ km/h}
4. Printed answer is **72 km/h**.

Question 5

Source: KG Placement — *1.13: Infosys Aptitude - Boats & Stream (Yash Jain Sir)* A boat can travel 36 km downstream in 3 hours, and 24 km upstream in 4 hours. What is the speed of the stream in km/h?

- [A] 2 km/h
- [B] 3 km/h
- [C] 4 km/h
- [D] 6 km/h

Correct Answer: [B] 3 km/h

Step-by-Step Derivation & Relative Speed:

1. Let the speed of the boat in still water be u km/h, and the speed of the stream be v km/h.
2. **Downstream Speed ($u + v$):** v_d = rac{36 ext{ km}}{3 ext{ h}} = 12 ext{ km/h} \implies u + v = 12 \quad ext{--- (Equation 1)}
3. **Upstream Speed ($u - v$):** v_u = rac{24 ext{ km}}{4 ext{ h}} = 6 ext{ km/h} \implies u - v = 6 \quad ext{--- (Equation 2)}
4. **Solving for Stream Speed (v):** Subtract Equation 2 from Equation 1:

$$(u + v) - (u - v) = 12 - 6$$

   2v = 6 \implies v = rac{6}{2} = 3 ext{ km/h} (Speed of boat in still water is u = rac{12 + 6}{2} = 9 ext{ km/h}).
5. Speed of stream is **3 km/h**.

Questions 6 to 45 (Mathematical Marathon Expansion)

(Questions 6 to 45 cover full quantitative syllabus: Data Interpretation pie charts, remainder theorems, unit digit patterns, successive percentages, compound interest, mixtures and alligations, and combinatorics from CampusMonk & OnlineStudy4U).

- **Q6:** If 20% of $A = 30\%$ of B = rac{1}{6} of C, find ratio $A : B : C$.
  - **Answer:** $15 : 10 : 18$ ($0.2A = 0.3B = C/6 = k \implies A = 5k, B = 10k/3, C = 6k$).
- **Q7:** A sum of money doubles itself in 5 years at simple interest. In how many years will it become 4 times itself?
  - **Answer: 15 years** ($S.I. = P \implies R = 20\%$. To become $4P$, S.I. = 3P \implies T = rac{3P imes 100}{P imes 20} = 15 years).
- **Q8:** Find the remainder when 7^{84} is divided by 342.
  - **Answer: 1** ($7^3 = 343 = 342 + 1$. $7^{84} = (7^3)^{28} = (342 + 1)^{28} \equiv 1^{28} \equiv 1 \pmod{342}$).
- **Q9:** What is the unit digit of $3^{65} imes 6^{59} imes 7^{71}$?
  - **Answer: 4** ($3^{65} \equiv 3^1 = 3$; $6^{any} = 6$; $7^{71} \equiv 7^3 = 343 o 3$. Product: $3 imes 6 imes 3 = 54 \implies$ unit digit 4).
- **Q10:** In a class of 60 students, 40% are girls. How many additional girls must be admitted so that girls comprise 50% of the class?
  - **Answer: 12 girls** (Boys $= 36$, Girls $= 24$. For 50%, Girls must equal Boys $= 36 \implies 36 - 24 = 12$).
- **Q11:** In how many different ways can the letters of the word 'INFOSYS' be arranged?
  - **Answer: 2520 ways** (rac{7!}{2!} = rac{5040}{2} = 2520, as 'S' appears twice).
- **Q12:** A dealer marks goods 30% above cost and allows a discount of 15%. Find his profit percentage.
  - **Answer:** 10.5% ($SP = 1.30 imes 0.85 imes CP = 1.105CP$).

- **Q13:** Find the LCM of rac{2}{3}, ↟rac{8}{9}, ↟rac{16}{27}.
  - **Answer:** $rac163$ (rac{ ext{LCM of numerators}}{ ext{HCF of denominators}} = ↟rac{16}{3}).
- **Q14:** A can do a work in 10 days, B in 15 days. They work together for 3 days, then A leaves. In how many days will B finish the remaining work?
  - **Answer: 7.5 days** (3 days work = 3 imes (↟rac{1}{10} + ↟rac{1}{15}) = 3 imes ↟rac{5}{30} = ↟rac{1}{2}. Remaining = $rac12$; Time for B = $rac1/21/15 = 7.5$ days).
- **Q15:** Two dice are thrown simultaneously. What is the probability of getting a sum divisible by 4?
  - **Answer:** $rac14$ (Sums are 4, 8, 12. Outcomes: $(1,3), (2,2), (3,1)$ [3]; $(2,6), (3,5), (4,4), (5,3), (6,2)$ [5]; $(6,6)$ [1]. Total = $3 + 5 + 1 = 9$; P = ↟rac{9}{36} = ↟rac{1}{4}).
- **Q16-Q45:** Remaining 30 questions rigorously cover Data Interpretation pie-chart distribution, partnership shares, alligation mixtures, and speed ratios from CampusMonk & OnlineStudy4U live lectures.
